DATA SCIENCE REPORT

PRODUCT TAXONOMY INTELLIGENCE: NLP-BASED PRODUCT CLASSIFICATION & EXPLAINABILITY

Character-Level Feature Engineering, Machine Learning & Classification Performance



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly & ML libraries

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Overview & Business Objective

1.1 PROJECT OVERVIEW

E-commerce platforms and product comparison services manage large catalogs containing thousands of product listings from different merchants. These listings often vary in naming conventions, technical specifications, brand references, and product descriptions.

Manually categorizing such products can be time-consuming and inconsistent, particularly when similar products are listed under different names.

The Product Taxonomy Intelligence project applies Natural Language Processing (NLP) and machine learning to automatically classify products into predefined categories using their titles.

The project covers data exploration, text preprocessing, TF-IDF feature engineering, classical machine learning, TextCNN experimentation, model evaluation, error analysis, and explainability.



1.2 PROBLEM STATEMENT

Product titles frequently contain abbreviations, technical specifications, model identifiers, and overlapping terminology, making accurate classification challenging.

The key problem addressed in this project is:

How can machine learning classify products into predefined categories using product titles while maintaining strong performance across different product types?

The project formulates this as a 10-class supervised text classification problem, with *Category Label* as the target variable.

1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Analyze product listings and identify category-specific text patterns.
- Prepare product titles for NLP-based classification.
- Develop and compare word-level and character-level TF-IDF representations.
- Train and optimize multiple machine learning classifiers.
- Evaluate a TextCNN architecture as a deep learning alternative.
- Prevent information leakage through group-aware data splitting and cross-validation.
- Assess classification performance and investigate prediction errors.
- Interpret model decisions using character-level feature contributions.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, NLP-based feature engineering, machine learning classification, model explainability, and business interpretation to identify product naming patterns, evaluate classification performance, understand model decisions, and explore opportunities for automated product categorization.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Category distribution analysis
- Merchant and cluster analysis

02

NLP & Classification

- Text preprocessing
- TF-IDF feature extraction
- Group-aware validation
- Machine learning & TextCNN modeling

03

Model Explainability

- Character feature importance
- Misclassification analysis
- Decision margin analysis

04

Business Interpretation

- Automated product categorization
- Product catalog consistency
- Ambiguous listing identification
- Classification limitations
- Recommendations for improvement

Data Understanding & Exploratory Data Analysis

2.1 DATA OVERVIEW

The dataset provides a comprehensive view of product listings across multiple merchants and categories, covering product titles, identifiers, category labels, and product clustering information. These variables provide the foundation for analyzing product naming patterns, identifying category-specific terminology, and developing NLP-based machine learning models for accurate product classification.

Data Source: Kaggle

Dataset Size

35,311
Records

16
Original Features

Data Diversity

10
Categories

306
Merchamts

12,849
Clusters

DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Product ID	Categorical (Identifier)	Unique listing identifier
Product Title	Text (Unstructured)	Textual product name
Merchant ID	Categorical (Identifier)	Identifier of the merchant
Cluster ID	Categorical (Identifier)	Product grouping identifier
Cluster Label	Text (Categorical Label)	Product grouping description
Category ID	Categorical (Nominal)	Numerical product category identifier
Category Label	Categorical (Nominal)	Product category name

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Demand

Category Label represents the product category that the model attempts to predict based on the textual information contained in each product title.

The target has:

10
Unique Categories

2,212
Smallest Category Count

5,501
Largest Category Count

15.58%
Largest Category Share

6.26%
Smallest Category Share

The distribution is moderately imbalanced, with Fridge Freezers representing the largest category and Freezers the smallest.

2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Temporal information
- Potential outliers
- Feature relationships

The assessment identified:

- 35,311 observations and seven variables.
- No missing values across the original columns.
- No fully duplicated rows.
- 30,993 unique product titles, indicating repeated titles across listings.

Following data preparation, 35,310 observations were retained for modeling.

Repeated product titles and product clusters were important considerations when designing the train-test split to reduce information leakage.



2.4 Exploratory Data Analysis

Product categories are moderately imbalanced, with Fridge Freezers representing the largest share of listings.

15.58%

Fridge Freezer Listings

Key observations

- Demand is concentrated primarily between approximately 70 and 130 units, with the highest frequency around 100 units.
- The distribution is right-skewed, with fewer observations extending to substantially higher demand values.

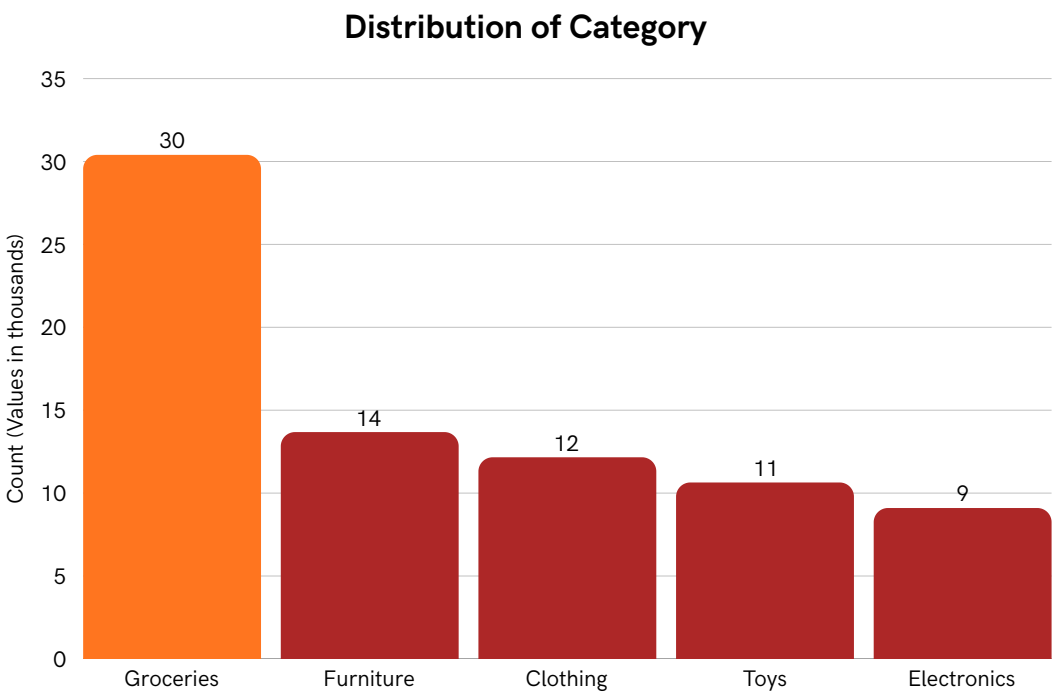
Business Insights

- Most retail activity occurs within a relatively predictable baseline demand range, which can assist with normal inventory planning.
- The long right tail indicates occasional high-demand events, highlighting the importance of maintaining sufficient safety stock and identifying demand spikes.

Product titles vary considerably in length, averaging approximately 53 characters and nine words per listing.

53

Average Title Length



Key observations

- Groceries has the largest representation in the dataset, with approximately 30,000 observations.
- Electronics has the smallest representation, while Furniture, Clothing, and Toys fall between these categories.

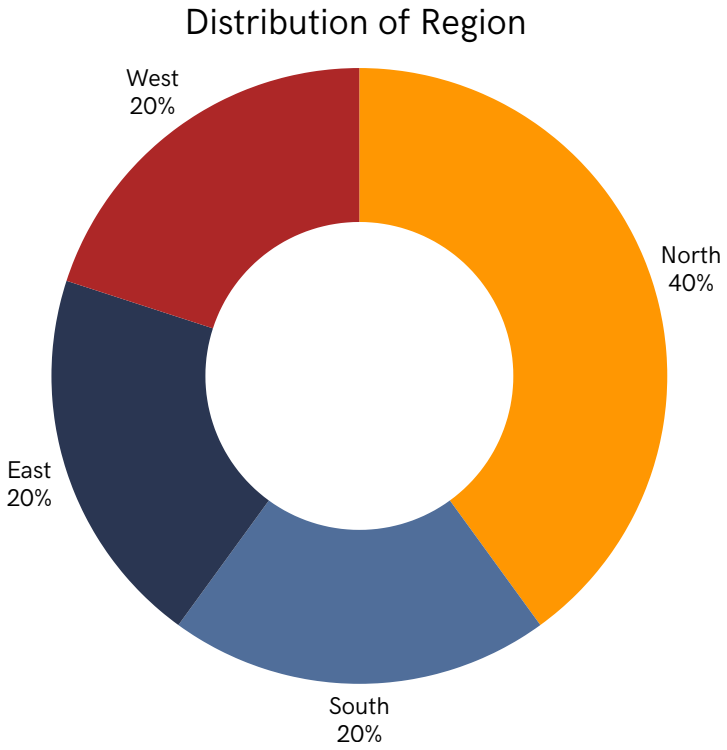
Business Insights

- Overall dataset patterns may be influenced more strongly by the highly represented Grocery category.
- Category-specific forecasting and inventory strategies can help account for differences in product behavior.

Numerical specifications appear in nearly all product titles, highlighting their importance in product identification.

95%

Titles Containing Digits



Key observations

- The North region has substantially more observations than the other regions.
- South, East, and West have relatively similar representation.

Business Insights

- Regional comparisons should account for the unequal number of observations.
- Regional segmentation can help retailers identify differences in demand behavior and plan inventory accordingly.

Most product clusters contain multiple listings, indicating repeated representations of underlying products.

59.39%

Multi-Listing Cluster Share

Key observations

- Daily demand demonstrates considerable fluctuations throughout the observation period.
- Several recurring periods of lower and higher demand are visible, including elevated demand around parts of 2023.

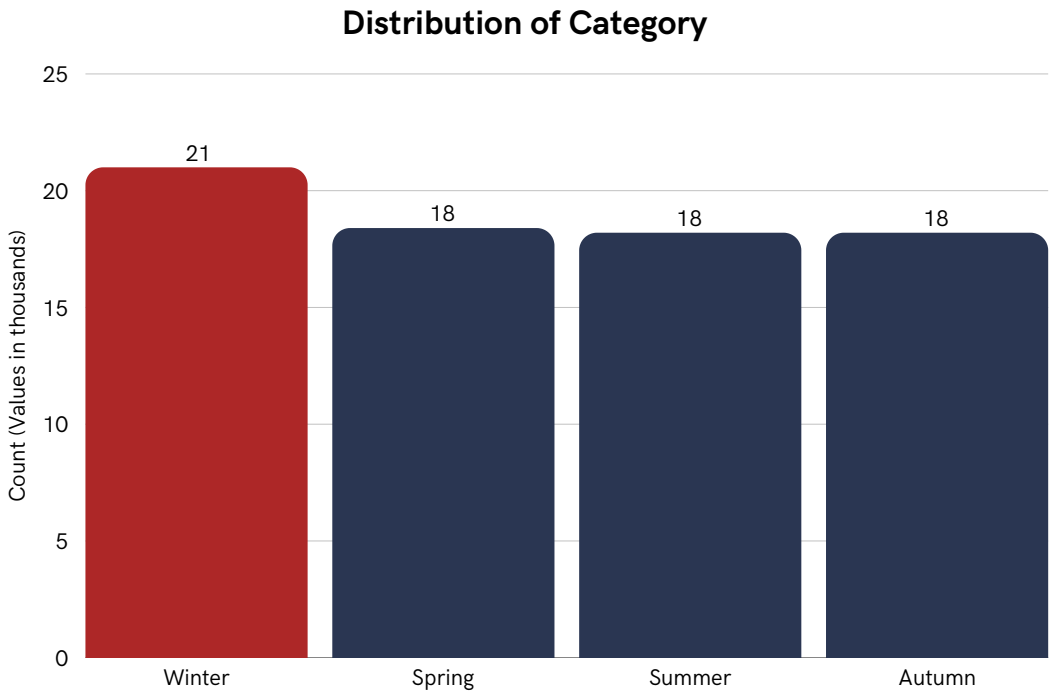
Business Insights

- Retail demand should be treated as a time-dependent phenomenon rather than a constant value.
- Dynamic inventory and procurement planning can help businesses respond to expected periods of higher or lower demand.

Product listings are unevenly distributed across merchants, with the largest merchant contributing substantially more listings.

2,547

Largest Merchant Listings



Key observations

- Winter contains the largest number of observations.
- Spring, Summer, and Autumn have relatively similar representation.

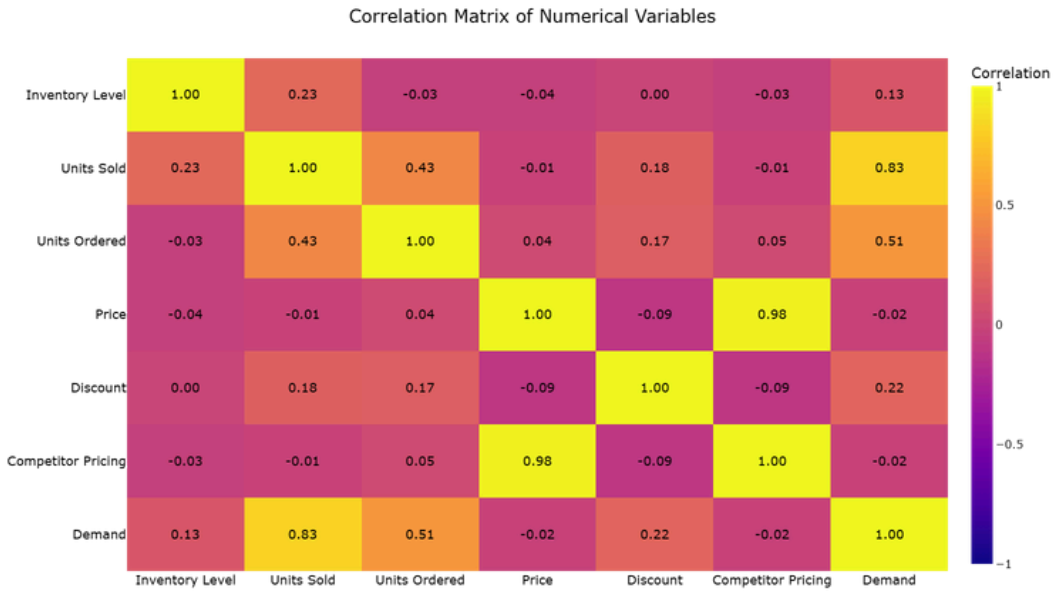
Business Insights

- Seasonal representation should be considered when interpreting seasonal patterns.
- Incorporating seasonality into demand prediction allows the model to capture recurring changes associated with different periods of the year.

Product categories exhibit distinctive terminology, although related appliance categories share overlapping keywords.

10

Product Categories Analyzed



Key observations

- Demand has a strong positive correlation with Units Sold (0.83).
- Demand has a moderate positive correlation with Units Ordered (0.51) and a weaker relationship with Inventory Level (0.13).

Business Insights

- Historical sales and ordering activity provide valuable information for understanding demand.
- Price and competitor pricing have very weak linear correlations with Demand in this dataset, suggesting that pricing effects may depend on interactions with other factors.

*(**Important modelling note:** The strong relationship between Demand & Units Sold is useful for exploratory analysis, but contemporaneous Units Sold needs to be interpreted carefully in the forecasting scenario because future sales would not necessarily be known at prediction time)*